

DECISION MEMO:

Inventory Optimization & Revenue Recovery Plan

To: Supply Chain Operations Committee

From: Vinita Chaudhari

Date: March 10, 2026

Subject: Transition to Statistical Replenishment to Recover Lost Revenue

1. Goal & KPIs

Comment: Defines the "North Star" metrics and strategic objectives for the project.

- **Primary Goal:** To eliminate the critical systemic stockout rate (identified at 99.2%) by transitioning from reactive manual ordering to an automated Statistical Reorder Point (ROP) policy.
 - **KPI 1: Revenue Recovery:** Targeted recovery of **₹6.5 Million** in lost sales over a 30-day horizon across 18 store locations.
 - **KPI 2: Stockout Rate:** Reduction of "Out of Stock" instances for high-velocity SKUs to achieve a 95% service level.
 - **KPI 3: Operational Efficiency:** Implementation of a standardized recommended_order_qty to minimize shipment frequency and logistics overhead.
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2. Data & Methodology

Comment: Details the Data Architecture and ETL workflow executed in etl_pipeline.py.

- **Data Architecture:** Integrated multi-source data including Product Metadata (JSON), Daily Sales/Inventory (CSV), and Purchase Orders to create a Single Source of Truth.
 - **ETL Process:** * **Normalization:** Automated cleaning of headers and date formats using Python (Pandas).
 - **Feature Engineering:** Calculated stockout_flag (binary indicator) and days_of_cover to identify points of failure.
 - **Demand Smoothing:** Applied a 28-day rolling average to account for sales volatility.
 - **Replenishment Policy:** Designed a Reorder Point (ROP) system using a **95% Service Level ($Z=1.645$)**, factoring in individual SKU demand standard deviation and supplier lead times.
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3. Key Insights (Analysis Findings)

Comment: Data-driven discoveries extracted from the analysis.ipynb and fact_sales datasets.

1. **Critical Availability Gap:** The network-wide stockout rate is **99.2%**, meaning nearly all revenue-generating products are currently unavailable for purchase.
 2. **Daily Financial Leakage:** The organization is losing an estimated **₹217,761 per day** in unrealized revenue due to empty shelves.
 3. **Category Concentration:** **Grocery and Snacks** are the most affected departments, representing over 45% of the total revenue recovery potential.
 4. **Lead Time Neglect:** Analysis shows supplier lead times range from 2 to 14 days, yet current ordering does not adjust "Safety Stock" to compensate for these variances.
 5. **High-Velocity Volatility:** 12% of SKUs exhibit high demand variance, requiring significantly higher buffer stock than currently maintained to avoid sales loss.
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4. Forecast Performance & Chosen Approach

Comment: Validates the forecasting logic displayed in View 2 of the Dashboard.

- **Methodology:** A **Baseline Demand Forecast** was calculated using a historical 28-day mean to capture recent consumer trends.
 - **Approach:** By comparing True Demand (the sales we *could* have made) against Censored Sales (zero sales because of zero stock), we identified that current stock levels are insufficient for 98% of the product catalog.
 - **Model Selection:** We moved away from simple Min/Max levels to a **Dynamic Demand Forecast** that automatically updates as sales trends change, visible in the "Actual vs. Forecast" dashboard view.
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5. Replenishment Plan (Operational Execution)

Comment: Explains the logic behind the "Replenishment Planner" in View 4.

- **The Trigger:** An automated Purchase Order is generated the moment On_Hand_Close falls below the calculated Reorder_Point.
 - **The Math:** $\$ROP = (\text{Average Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}\$$.
 - **Execution Example: * SKU0001 (Personal Care):** Daily Demand is ~0.27 units. With a 3-day lead time and calculated Safety Stock of 2.97 units, the **Order Trigger is set at 4 units**.
 - **Action:** When stock reaches 4, the system automatically recommends an order of **5 units** to return to the target stock level.
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6. 30-Day Impact Estimate

Comment: Projections based on the implementation of the final_replenishment_policy.csv.

- **Revenue Impact:** Total 30-day projected revenue recovery is **₹6,532,841**.
 - **Inventory Health:** Transitioning to this policy will result in a **98% reduction in stockouts**, ensuring that high-margin items are always available.
 - **Assumption:** The estimate assumes supplier lead times remain within calculated averages and that Recommended POs are approved within 24 hours of generation.
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7. Risks, Limitations & Next Steps

Comment: Future-proofing the strategy.

- **Risks:** Sudden spikes in demand (e.g., viral trends) may exceed a 95% service level; supplier delays beyond 14 days still pose a stockout risk.
- **Limitations:** The current model does not account for physical warehouse/shelf volume constraints.
- **Next Steps:**
 1. **Pilot Program:** Implement the "Replenishment Planner" list in the three stores with the highest revenue leakage (ST001, ST012, ST018).
 2. **Promotion Integration:** Incorporate the promo_flag into future forecasts to "pre-stock" items before marketing campaigns begin.
 3. **Automation:** Transition the Python ETL pipeline to a cloud environment (e.g., Azure/AWS) for real-time dashboard updates.

Final Recommendation:

The data confirms that current inventory management is purely reactive. Immediate adoption of the **Statistical Replenishment Policy** is required to stop the loss of **₹6.5M** in monthly revenue and restore operational stability.